

8086 Microprocessor & PLC Practice Questions – Solved Notes

Question 1: PUSHA Instruction

Explain what happens when the **PUSHA** instruction executes. Show clearly the physical memory addresses where each register is stored. Assume **SP = 0006H** and **SS = A000H** before execution.

Given:

- **SP = 0006H**
- **SS = A000H**

PUSHA pushes registers in the following order:

$$AX \rightarrow CX \rightarrow DX \rightarrow BX \rightarrow \textbf{Original SP} \rightarrow BP \rightarrow SI \rightarrow DI$$

Initial physical address:

$$A000H \times 16 + 0006H = A0006H$$

Register	SP after push	Physical Address
AX	0004H	A0004H
CX	0002H	A0002H
DX	0000H	A0000H
BX	FFFEH	9FFFEH
SP (orig)	FFFCH	9FFFCH
BP	FFFAH	9FFFAH
SI	FFF8H	9FFF8H
DI	FFF6H	9FFF6H

Final SP = FFF6H

Question 2: CWD and IDIV

What will be the contents of the **DX:AX** register pair after execution of the instructions **CWD** and **IDIV CX** if **CX = FF8DH** and **AX = FBE4H**?

AX is negative, so CWD sign-extends AX into DX.

$$DX = FFFFH$$

Dividend:

$$DX : AX = FFFFFBE4H = -1052$$

Divisor:

$$CX = FF8DH = -115$$

$$\frac{-1052}{-115} = 9 \quad \text{remainder} = -17$$

$$AX = 0009H, \quad DX = FFEFH$$

Question 3: Address Calculation

Suppose DS = 1300H, SS = 1400H, BP = 1500H, and SI = 0100H. Determine the physical address accessed by each instruction and mention the addressing mode (assume real mode operation):

1. MOV AL, [BP+SI-200H]
2. MOV AL, [SI-0100H]

(i) MOV AL, [BP+SI-200H]

$$1500H + 0100H - 0200H = 1400H$$

Uses SS segment:

$$14000H + 1400H = 15400H$$

Addressing Mode: Based Indexed with Displacement

(ii) MOV AL, [SI-0100H]

$$0100H - 0100H = 0000H$$

Uses DS segment:

$$13000H$$

Addressing Mode: Indexed with Displacement

Question 4: FLAGS and Segment Range

Determine the values of the FLAG register bits after executing:

MOV AX, 00A0H

MOV BX, 0005H

SUB AX, BX

Also determine the physical address range of a segment register given initial address 0C51:CDD0H.

$$AX - BX = 009BH$$

Flag	Value
CF	0
ZF	0
SF	0
OF	0
AF	1
PF	1

$$\text{Base Address} = 0C51H \times 16 = 0C510H$$

$$\text{Segment Range} = 0C510H \text{ to } 1C50FH$$

Question 5: LEA and LDS

Explain the operation of the following instructions and show their effect on the stack pointer:

1. LEA OP1, OP2
2. LDS BX, [4326]

LEA:

- Loads effective address
- SP is not affected

LDS:

- Loads offset into BX
- Loads segment into DS
- SP is not affected

Question 6: Array Summation Program

Write an assembly language program to add two arrays, each of size 100 words, and store the result on top of the data in the second array. Use LODSW and STOSW.

```
MOV CX, 100          ; Loop counter
LEA SI, ARRAY1       ; Source array address
LEA DI, ARRAY2       ; Destination array address

SUMLOOP:
    LODSW            ; AX = [SI], SI += 2
    ADD AX, [DI]     ; Add second array element
    STOSW            ; Store result in ARRAY2
    LOOP SUMLOOP

HLT
```

Question 7: PLC Control System

A pump must turn ON only when a high-temperature switch AND a low-pressure switch are ON. An emergency stop must turn the pump OFF at any time.

(a) Field Devices

- Inputs: High Temp Switch, Low Pressure Switch, Emergency Stop
- Output: Pump Motor

(b) Hardwired Connections

120 VAC inputs connected to PLC input module; output module controls motor contactor.

(c) Ladder Logic

High Temp AND Low Pressure in series, Emergency Stop as normally-closed contact.

(d) Installation and RUN

- Wire with power OFF
- Assign I/O addresses
- Download program
- Switch PLC to RUN mode

(e) Modified Control

Manual pushbutton operates at any pressure but only when temperature is above the set value.